

MEASURING EARTHQUAKES

Scientists measure earthquake vibrations with an instrument called a seismometer. In various forms, the seismometer has been in use for thousands of years.

1855
Italian physicist Luigi Palmieri designs a seismometer that can record the direction, intensity, and duration of earthquakes too small for humans to notice.



1902
Italian scientist Giuseppe Mercalli invents a scale for measuring earthquakes based on observation of effects.

1925
Americans Harry Wood and James Anderson's seismometer is precise enough to be used ten years later for the Richter scale.



Seismograph, recorded 1920s

AROUND 500,000 EARTHQUAKES ARE RECORDED BY INSTRUMENTS EVERY YEAR

1700

1703
French inventor Jean de Hautefeuille builds a basic seismometer by filling a bowl with mercury and noting the amount and direction of spill during an earthquake.

1751
Italian teacher Andrea Bina uses a pendulum with a pointer to trace movement in the sand below during an earthquake.



Bina's pendulum

1880
While working in Japan, British geologist John Milne develops the first accurate seismometer.

1907
German physicist Emil Wiechert builds a machine that records an earthquake using an oscillating pendulum.



Wiechert's pendulum

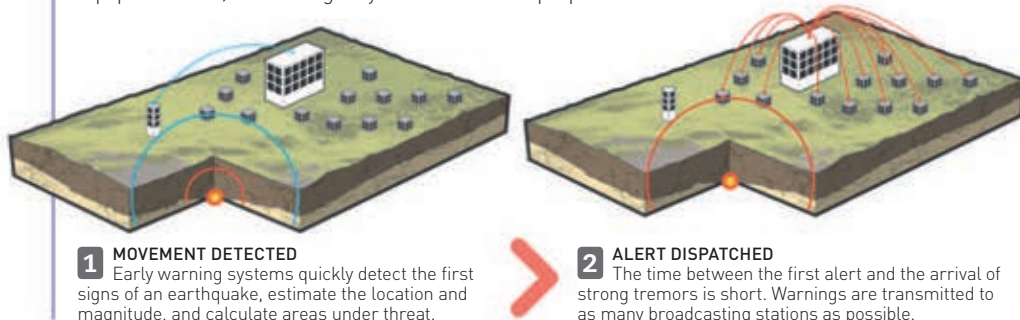
1934
American seismologist Charles Richter develops a widely used scale that measures the energy released by an earthquake.

1979-PRESENT
The moment magnitude scale is introduced as a more accurate version of the Richter scale.

2015

EARLY WARNING SYSTEMS

Early warning systems act to protect people, animals, and property by alerting people of incoming seismic waves from an earthquake. This gives people time to take cover, businesses and power stations time to make equipment safe, and emergency services time to prepare for action.



EARTHQUAKE DRILL

One simple emergency drill has been proven to reduce injuries from earthquakes: Drop, Cover, Hold On. This is because most injuries come from falling objects such as lamps and glass, rather than from building collapse.



- 1 DROP**
Drop to the ground immediately to protect yourself as much as possible where you are.
- 2 COVER**
Take cover under a sturdy desk or table if possible; if not, move to the corner of the room.
- 3 HOLD ON**
Hold on to the desk or table while covering your head and neck with your arms.

RESISTING EARTHQUAKES

While no structure can be guaranteed completely safe from earthquake damage, the buildings listed here have proven to be very resistant to massive ground shakes.

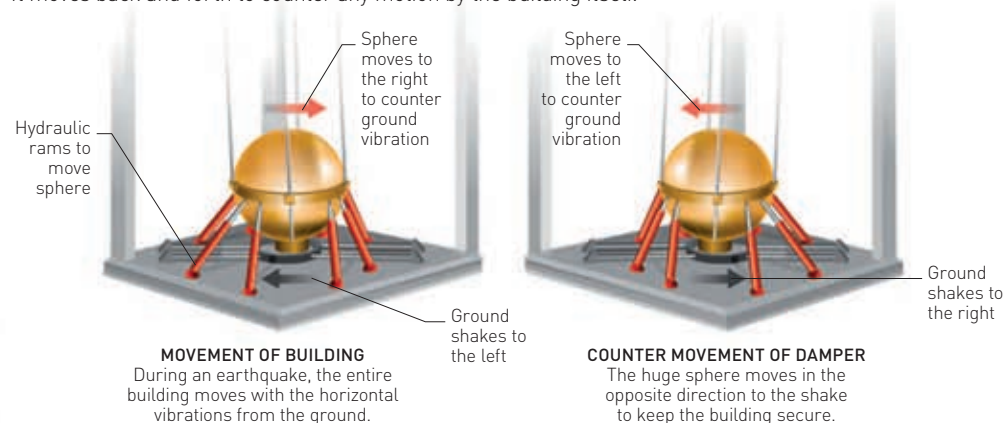
- CHECHEN ITZA, MEXICO**
The Mayan pyramid of El Castillo at Chechen Itza is very strong as it has a base much broader than its summit.
- TOMB OF CYRUS, IRAN**
Built in 400 BCE, this uses "base-isolation" to survive shakes: its base moves independently of its foundations.
- YOKOHAMA LANDMARK TOWER, JAPAN**
This skyscraper has a mass damper system, sits on rollers, and is made from flexible materials.
- TRANSAMERICA PYRAMID, USA**
Rising to 260 m (853 ft), this skyscraper in San Francisco has foundations that reach 16 m (52 ft) into the ground.
- TAIPEI 101, TAIWAN**
Stretching twice as high as the Transamerica Pyramid, this relies on a huge mass damper to resist movement.



TAIPEI 101

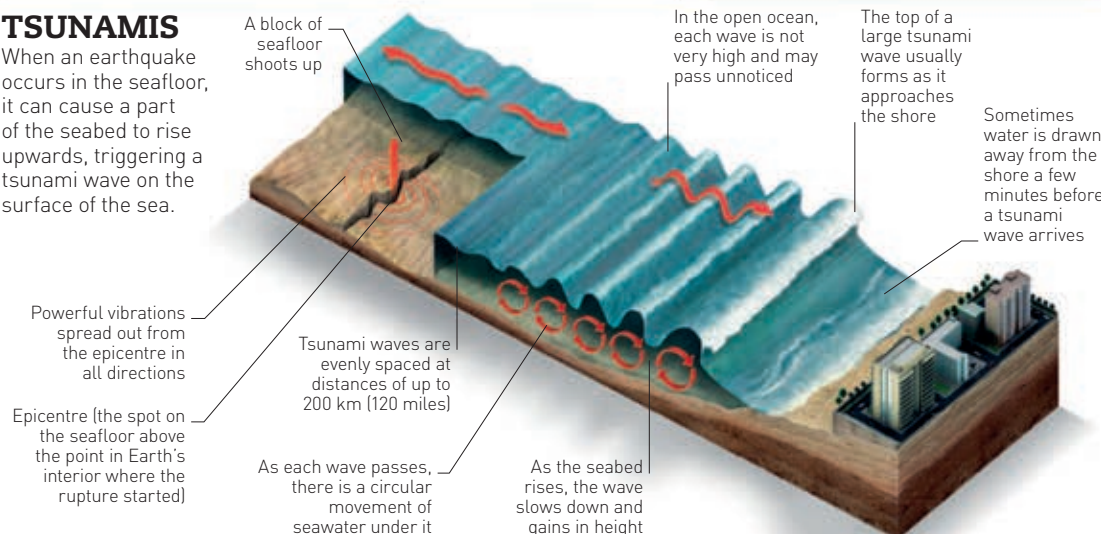
MASS DAMPER

One way to help skyscrapers cope with an extreme ground shake is to install a mass damper – a huge steel sphere – at the centre, suspended by cables. It moves back and forth to counter any motion by the building itself.



TSUNAMIS

When an earthquake occurs in the seafloor, it can cause a part of the seabed to rise upwards, triggering a tsunami wave on the surface of the sea.



MAJOR TSUNAMIS

Tsunamis are assessed according to the size of their waves, how many occur in one event, how far they come on shore, and how much damage they cause.

- 1 SUMATRA, INDONESIA, 26 DECEMBER 2004**
This tsunami's waves reached 50 m (164 ft) and killed more than 227,000 people, affecting 14 countries.
- 2 NORTH PACIFIC COAST, JAPAN, 11 MARCH 2011**
Travelling at 800 km/h (497 mph), the 10 m (33 ft) high waves of this tsunami forced 450,000 people from their homes.
- 3 PORTUGAL, 1 NOVEMBER 1755**
Set off by an 8.5-magnitude earthquake, this tsunami hit Portugal, Morocco, and Spain with waves 30 m (98 ft) high.
- 4 KRAKATOA, INDONESIA, 27 AUGUST 1883**
Caused by the eruption of the Krakatoa Caldera volcano, this tsunami created multiple waves reaching 37 m (121 ft) high.
- 5 ENSHUNADA SEA, JAPAN, 20 SEPTEMBER 1498**
Waves from this tsunami were powerful enough to cross a section of land separating Lake Hamana from the sea.